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Report

Laboratory Work No. 2

Cryptography

Topic: Cryptanalysis of Monoalphabetic Ciphers

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1 Objective and scope

The aim of this laboratory was to recover an English plaintext encrypted with a monoalphabetic substitution cipher, without knowing the substitution key. The work was carried out manually using the Crypto Corner frequency-analysis activity [2]. It was a cryptanalysis exercise, with no programming implementation required.

The solution combined letter frequencies with repeated words, common letter combinations, grammatical context, and consistency across the passage. The working record contains 19 deduction steps; several steps establish more than one letter pair. This report follows that order and illustrates how successive substitutions make the message readable. The ciphertext and the final literal decryption are included in the appendices.

2 Theoretical background

2.1 Monoalphabetic substitution and its weakness

A monoalphabetic substitution replaces each plaintext letter with a fixed ciphertext letter throughout a message. For an alphabet $\mathcal{A}$ of 26 English letters, let π be a permutation of that alphabet. Encryption and decryption are

$$c_i = \pi(p_i), \quad p_i = \pi^{-1}(c_i). \quad (1)$$

A general substitution has $26! \approx 4.03 \times 10^{26}$ possible keys. This is far larger than the 25 nonzero shifts of the English Caesar cipher, but the size of the key space does not remove statistical structure. Every occurrence of a plaintext letter receives the same replacement. Repeated letters, repeated words, and frequency patterns survive under different labels.

The analysis therefore targets language structure rather than trying every key. If a plaintext letter appears n times, its ciphertext replacement also appears n times. When spaces and punctuation are retained, as in this cryptogram, they provide additional clues about word lengths, sentence boundaries, and possessive forms.

2.2 Letter-frequency analysis

The course material explains that letters occur at different rates in natural languages [1]. For a ciphertext containing N letters, the observed percentage of a symbol c is

$$f_C(c) = \frac{n_C(c)}{N} \times 100\%, \quad (2)$$

where $n_C(c)$ counts that symbol. Uppercase and lowercase forms are counted together. Spaces, punctuation, digits, and line breaks are excluded from N .

A sufficiently long ordinary passage often resembles the broad frequency profile of its language, although subject matter and writing style cause deviations. English is the relevant reference for this exercise. The Romanian frequencies in the handout illustrate the same language-dependent

principle, but they are not the reference used to solve this English passage.

Table 1: English reference frequencies from the supplied course material (percent)

Letter	Frequency	Letter	Frequency
A	8.17	N	6.75
B	1.49	O	7.51
C	2.78	P	1.93
D	4.25	Q	0.09
E	12.70	R	5.99
F	2.23	S	6.33
G	2.01	T	9.06
H	6.09	U	2.76
I	6.97	V	0.98
J	0.15	W	2.36
K	0.77	X	0.15
L	4.03	Y	1.97
M	2.41	Z	0.07

The most frequent ciphertext symbol is a candidate for plaintext **e**, but this is a hypothesis, not a rule. Matching the sorted frequency lists mechanically can produce incorrect substitutions. A few likely pairs constrain the key; they do not uniquely establish the entire permutation. Each proposal must also fit words and remain consistent with previously accepted pairs.

2.3 Word patterns, digraphs and trigraphs

A digraph is a pair of letters and a trigraph is a group of three. The supplied material lists common English digraphs such as TH, HE, AN, IN, ER, ON, RE, and ED. Useful trigraphs include THE, AND, THA, ENT, ION, and TIO. A recurring pattern such as **tQe** can therefore suggest the word **the** and the pair **Q** $\rightarrow$ **h**.

Other clues include the ordinary one-letter English words **a** and **i**, common doubled letters such as **ss**, **ee**, **tt**, **oo**, and **ff**, and word shapes that survive substitution. Once several letters are known, a pattern such as **teYt** strongly suggests **text**. Context then tests the proposal across all occurrences, rather than accepting it because it fits one location.

2.4 Method and limitations

The working method was to compare frequencies, propose a substitution, inspect the partially decoded passage, and accept or reconsider the proposal based on additional words. Each ciphertext letter must have one consistent plaintext value; two ciphertext letters cannot both represent the same plaintext letter under the assumed permutation.

Long messages usually provide more repeated evidence than short ones, but even a long passage can have unusual vocabulary. Proper names and missing spaces are especially poor starting points. Here, common words were solved first; names such as Khnumhotep became readable later. Human judgement guided this particular exercise. This does not imply that substitution ciphers cannot be attacked automatically; the report describes the manual method used in the laboratory.

3 Frequency evidence and working method

3.1 The supplied cryptogram

The supplied text contains **2,567 alphabetic characters** and uses **25 distinct ciphertext letters**. Ciphertext M is absent. Counting the supplied text reproduces every integer count in the frequency screenshot, providing a check that the report uses the same cryptogram as the exercise.

The activity on the Crypto Corner website supplied the frequency displays used during the exercise [2]. The four images below are the supplied laboratory screenshots. The tables show the ranked distributions; the charts show the same kind of information in alphabetic order. The letter substitutions and their justifications come from the student’s working notes [3].

V	W	N	X	G	T	I	P	Q	O	H	S	Z	F	C	U	D	J	A	R	K	L	Y	B	E	M
327	276	205	198	196	178	176	165	145	93	83	80	63	61	59	58	56	45	35	27	17	13	8	2	1	0
12.7	10.8	8.0	7.7	7.6	6.9	6.9	6.4	5.6	3.6	3.2	3.1	2.5	2.4	2.3	2.3	2.2	1.8	1.4	1.1	0.7	0.5	0.3	0.1	0.0	0.0

Figure 1: Ranked ciphertext frequencies: symbol, count, and rounded percentage

E	T	A	O	I	N	S	H	R	D	L	C	U	M	W	F	G	Y	P	B	V	K	J	X	Q	Z
12.7	9.1	8.2	7.5	7.0	6.7	6.3	6.1	6.0	4.3	4.0	2.8	2.8	2.4	2.4	2.2	2.0	2.0	1.9	1.5	1.0	0.8	0.15	0.15	0.10	0.07

Figure 2: Ranked English reference frequencies used for comparison

For example, ciphertext V occurs 327 times:

$$f_C(V) = \frac{327}{2567} \times 100\% \approx 12.74\%. \quad (3)$$

This agrees closely with the English reference for e, approximately 12.7%. Ciphertext W occurs 276 times, or approximately 10.75%, and was proposed as t. The screenshot displays these as 12.7% and 10.8%. Its rounded reference figures may differ slightly from the two-decimal values in Table 1.

Frequency order alone is insufficient. The third most frequent ciphertext symbol is N, with 205 occurrences. A strict rank match would associate it with English a, whereas the readable word to establishes $N \rightarrow O$. Ciphertext T, only sixth in the observed ranking, is the actual replacement for a.

3.2 Notation and consistency

All mappings in the report run from ciphertext to plaintext: $V \rightarrow e$ means that ciphertext V represents plaintext e. In intermediate extracts, unresolved ciphertext symbols are uppercase and recovered plaintext letters are lowercase. This convention avoids confusing an unsolved symbol with a letter that has already been identified.

The snapshots in Section 5 are reconstructed from the supplied ciphertext using the cumulative mappings in the recorded order. They are text illustrations, not additional screenshots of the website. This makes the progression internally consistent while preserving the original deduction logic.

3.3 Visual comparison of the distributions

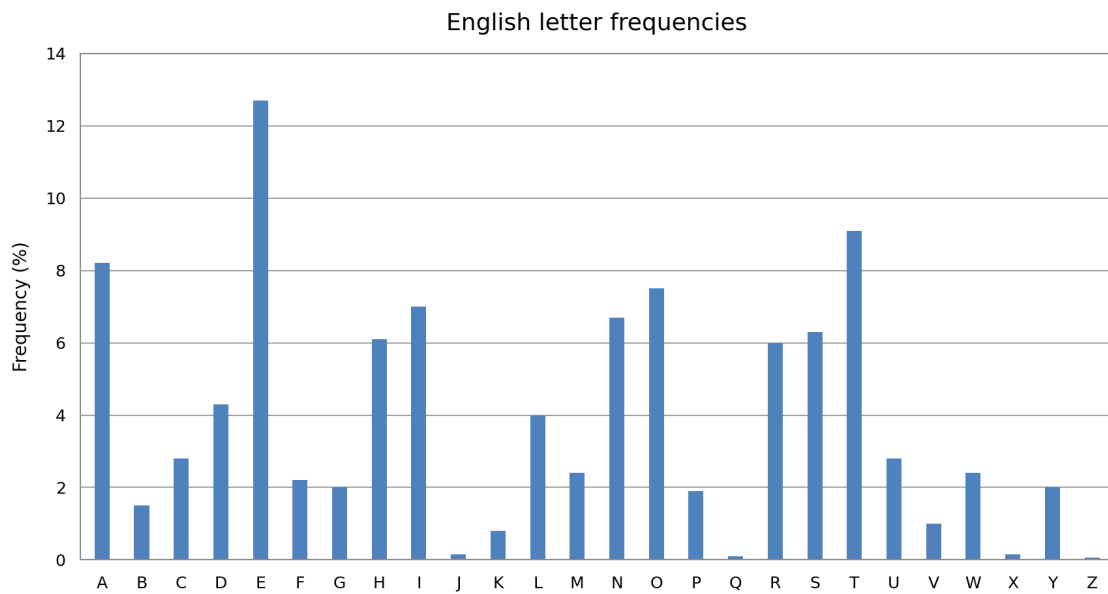


Figure 3: English reference profile: prominent peaks include E and T

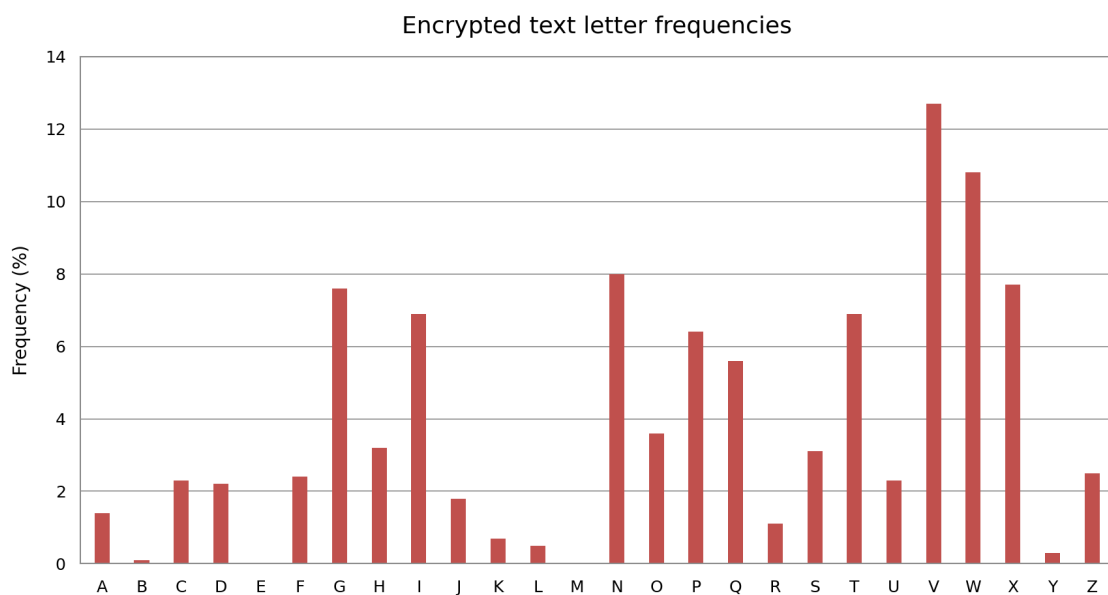


Figure 4: Ciphertext profile: the highest peaks occur at V and W

The cipher relabels the letters of this particular passage; it does not force the passage to have exactly the average English distribution. Similar peaks suggest initial matches, while word evidence determines whether they are correct. The following reconstruction preserves that distinction between a statistical guess and a substitution supported by readable context.

4 Step-by-step recovery of the substitution

The following sequence follows the 19 steps recorded during the exercise. The first decisions use frequency; later decisions rely increasingly on words revealed by earlier replacements. A candidate was useful when it explained several occurrences without contradicting the existing mapping.

4.1 Steps 1–7: establishing common letters

1. **V → e: the strongest frequency match.** Ciphertext V has 327 occurrences, approximately 12.74% of all letters, and is the most frequent symbol. This closely matches the English reference for e, making it the initial hypothesis.
2. **W → t: the second major peak.** Ciphertext W occurs 276 times, approximately 10.75%. Its high rank suggests t, even though this passage contains more of it than the average English percentage. Together, the first two replacements expose many occurrences of tQe.
3. **Q → h: recognizing the.** The repeated trigraph tQe is naturally read as the. This yields h and supplies word-level support for the earlier choices of t and e.
4. **T → a: recognizing that.** The pattern thTt becomes that. Ciphertext T also occurs as a one-letter word, supporting a. The agreement between a word pattern and a grammatical clue is stronger than a frequency guess alone.
5. **P → s: recognizing these.** The pattern thePe suggests these. Ciphertext P occurs 165 times, approximately 6.43%, close to the English reference of 6.33% for s. The screenshot rounds the observed value to 6.4%.
6. **N → o: recognizing to.** The frequent two-letter pattern tN becomes to. The observed 205 occurrences, approximately 7.99%, are compatible with a common vowel. This decision demonstrates why frequency ranks must be checked against words: N is third in the ciphertext ranking but represents o, not a.
7. **X → i: recognizing it.** The pattern Xt is read as it. The observed frequency, 198 occurrences or approximately 7.71%, supports a common letter such as i. This also begins to reveal recurring short grammatical words elsewhere in the passage.

4.2 Steps 8–12: recovering phrases

8. **Z → m, I → r, G → n: a complete phrase.** The fragment the saZe IeasoG as becomes the same reason as. It supplies three mutually compatible replacements. Their frequencies remain plausible, but the readable phrase is the decisive evidence. The beginning of the message now contains on a and nearSF.
9. **O → d: matching both ends of a phrase.** The fragment Oesire to reaO becomes desire to read. The same unknown symbol occurs at the start of one word and the end of another; both positions support the same letter.
10. **F → y, S → l, J → g: the opening sentence.** The fragment on a daF nearSF 4,000 Fears aJo becomes “on a day nearly 4,000 years ago”. The repeated F is explained con-

sistently in day, nearly, and years. This step establishes g, which also resolves endings such as doing and writing.

11. **R → w, H → c: recognizing town called.** The phrase in a toRn Halled menet becomes in a town called menet. The ordinary words provide the evidence; identifying the proper name menet in advance is unnecessary.
12. **C → f: of and life.** The fragment story oC his lord's liCe becomes story of his lord's life. The same substitution solves a common preposition and a contextually appropriate noun, strengthening the consistency of the partial key.

4.3 Steps 13–19: resolving the remaining observed symbols

13. **L → k: recognizing knows.** The fragment world Lnows it becomes world knows it. This resolves an uncommon letter using a familiar phrase rather than its low frequency alone.
14. **D → u, A → b: recognizing scribe and out.** The fragment a master scriAe sketched oDt becomes a master scribe sketched out. The replacements also clarify words and names elsewhere, including about, tomb, and Khnumhotep.
15. **U → p: recognizing hieroglyphs.** The joined fragment thehieroglyUhs becomes thehieroglyphs. A word boundary can be restored for reading, giving “the hieroglyphs”, but the missing space is independent of the substitution itself.
16. **B → q: recognizing technique.** The word techniBue becomes technique. Ciphertext B occurs only twice, so context is more informative than a frequency ranking for this rare letter.
17. **K → v: recognizing even.** The fragment eKen the slightest becomes even the slightest. The same pair also resolves words such as developed and service.
18. **Y → x: recognizing text.** The phrase read the teYt becomes read the text. This provides a clear context for a relatively rare letter: ciphertext Y appears eight times.
19. **E → j: recognizing just.** The fragment instead of Eust writing becomes instead of just writing. Ciphertext E appears only once. This final contextual deduction resolves the last symbol actually present in the cryptogram.

5 Progressive decryption

Each sample shows the same opening three lines, so the effect of accumulating substitutions can be compared directly. Uppercase characters are unresolved ciphertext; lowercase characters are recovered plaintext. Line wrapping is adjusted for page width. The grouping follows the supplied progress record. The full 19-step sequence is explained in Section 4.

5.1 After steps 1 and 2

Only $V \rightarrow e$ and $W \rightarrow t$ are known. The repeated pattern tQe becomes visible.

```
NG T OTF GeTISF 4,000 FeTIP TJN, XG T tNRG HTSSe0 ZeGet LQDCDANI0eIXGJ tQe
tQXG IXAANG NC tQe GXSe, T ZTPteI PHIXAe PLethQe0 NDt tQeQXeINJSFUQP tQTt
tNSO tQe PtNIF NC QXP SNIO'P SXCe— TGO XG PN ONXGJ QeNUeGe0 tQe IeHNI0e0
QXPtNIF NC HIFUtNSNJF. QXP RTP Gnt T PFPteZ NC
```

5.2 After step 3

The substitution $Q \rightarrow h$ resolves the , giving a reliable anchor at several locations.

```
NG T OTF GeTISF 4,000 FeTIP TJN, XG T tNRG HTSSe0 ZeGet LhDCDANI0eIXGJ the
thXG IXAANG NC the GXSe, T ZTPteI PHIXAe PLethHe0 NDt thehXeINJSFUhP thTt
tNSO the PtNIF NC hXP SNIO'P SXCe— TGO XG PN ONXGJ heNUeGe0 the IeHNI0e0
hXPtNIF NC HIFUtNSNJF. hXP RTP Gnt T PFPteZ NC
```

5.3 After step 4

The substitution $T \rightarrow a$ reveals the one-letter word a and the word $that$.

```
NG a 0aF GeaISF 4,000 FeaIP aJN, XG a tNRG HaSSe0 ZeGet LhDCDANI0eIXGJ the
thXG IXAANG NC the GXSe, a ZaPteI PHIXAe PLethHe0 NDt thehXeINJSFUhP that
tNSO the PtNIF NC hXP SNIO'P SXCe— aGO XG PN ONXGJ heNUeGe0 the IeHNI0e0
hXPtNIF NC HIFUtNSNJF. hXP RaP Gnt a PFPteZ NC
```

5.4 After steps 5 and 6

The additional pairs $P \rightarrow s$ and $N \rightarrow o$ reveal common word fragments and make the sentence structure easier to follow.

```
oG a 0aF GeaISF 4,000 FeaIs aJo, XG a toRG HaSSe0 ZeGet LhDCDAoI0eIXGJ the
thXG IXAAoG oC the GXSe, a ZasteI sHIXAe sLetHHe0 oDt thehXeIoJSFUhs that
toSO the stoIF oC hXs SoIO's SXCe— aGO XG so 0oXGJ heoUeGe0 the IeHoI0e0
hXstoIF oC HIFUtoSoJF. hXs Ras Got a sFsteZ oC
```

5.5 After steps 7 and 8

Adding i , m , r , and n makes $on a, master, and his$ readable. The broader passage supplies the phrase $the same reason as$ used in step 8.

```
on a 0aF nearSF 4,000 Fears aJo, in a toRn HaSSe0 menet LhDCDAorOerinJ the
thin riAAon oC the niSe, a master sHriAe sLetHHe0 oDt thehieroJSFUhs that
toSO the storF oC his SorO's SiCe— anO in so 0oinJ heoUene0 the reHorOe0
historF oC HrFutoSoJF. his Ras not a sFstem oC
```

5.6 After steps 9 and 10

The opening now reads “on a day nearly 4,000 years ago”. The new substitutions also reveal words such as **nile** and **story**.

on a day nearly 4,000 years ago, in a toRn Halled menet LhDCDAordering the
thin riAAon oC the nile, a master sHriAe sLetHhed oDt thehieroglyUhs that
told the story oC his lord's liCe— and in so doing heoUened the reHorded
history oC HryUtology. his Ras not a system oC

5.7 After steps 11 and 12

The substitutions for **w**, **c**, and **f** expose “in a town called” and “story of his lord’s life”.

on a day nearly 4,000 years ago, in a town called menet LhDfDAordering the
thin riAAon of the nile, a master scriAe sLetched oDt thehieroglyUhs that
told the story of his lord's life— and in so doing heoUened the recorded
history of cryUtology. his was not a system of

5.8 After steps 13–15

The remaining letters in **scribe**, **sketched**, and **hieroglyphs** are resolved. The opening sample happens to contain none of the symbols still awaiting steps 16–19; the rest of the passage still needs them.

on a day nearly 4,000 years ago, in a town called menet khufubordering the
thin ribbon of the nile, a master scribe sketched out thehieroglyphs that
told the story of his lord's life— and in so doing heopened the recorded
history of cryptology. his was not a system of

5.9 After steps 16–19

The whole ciphertext is now readable. This opening sample is unchanged from the preceding one because the final four pairs are established elsewhere, in **technique**, **even**, **text**, and **just**.

on a day nearly 4,000 years ago, in a town called menet khufubordering the
thin ribbon of the nile, a master scribe sketched out thehieroglyphs that
told the story of his lord's life— and in so doing heopened the recorded
history of cryptology. his was not a system of

These samples illustrate why a short readable excerpt is not sufficient proof that the entire message has been solved. The final verification must cover every symbol and the complete passage.

6 Recovered key and final result

6.1 The recovered substitution alphabet

Table 2 records the direction used throughout the solution: ciphertext letters map to plaintext letters. The 19 steps identify 25 pairs because some steps establish several pairs at once.

Table 2: Ciphertext-to-plaintext substitution key

Ciphertext	A	B	C	D	E	F	G	H	I	J	K	L	M
Plaintext	b	q	f	u	j	y	n	c	r	g	v	k	z*
Ciphertext	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Plaintext	o	d	s	h	w	l	a	p	e	t	i	x	m

*The pair $M \rightarrow Z$ is inferred by elimination; ciphertext M does not occur.

All 25 observed ciphertext letters have distinct plaintext values. The only unused plaintext letter is Z; therefore, assuming a permutation of the full 26-letter alphabet, the remaining pair must be $M \rightarrow Z$. No occurrence in this passage directly tests that pair.

6.2 Verification of the result

The recovered mapping explains the full passage consistently, including the common words used to derive it and the less common names that emerged later. It also preserves the expected repeated-letter structures. For example, a doubled ciphertext symbol must decrypt to a doubled plaintext letter, rather than requiring a different interpretation in each position.

As a preparation check, the inverse mapping was applied to the literal decrypted text. It reproduced the supplied ciphertext exactly after case normalization, including its punctuation, spacing, and line breaks. This confirms consistent application of the recovered key; by itself, a round trip would not prove that a guessed key gives the intended English message. Here the readable text and the independent word clues provide that additional evidence. The 26 integer frequency counts also match the screenshot exactly, including the zero count for M.

6.3 Complete plaintext: reading version

The following version applies the recovered key and restores obvious missing word boundaries and sentence capitalization. Paragraph breaks are editorial. The stray bullet before **proliferated** is omitted. No letter sequence is changed relative to the literal decryption apart from capitalization; the original joined words remain available in Appendix B. The historical statements are part of the recovered passage, not separate findings established by this cryptanalysis exercise.

On a day nearly 4,000 years ago, in a town called Menet Khufu bordering the thin ribbon of the Nile, a master scribe sketched out the hieroglyphs that told the story of his lord's life—and in so doing he opened the recorded history of cryptology. His was not a system of secret writing as the modern world knows it; he used no fully developed code of hieroglyphic symbol substitutions. His inscription, carved about 1900 B.C. into the living rock in the main chamber of the tomb of the nobleman Khnumhotep II,

merely uses some unusual hieroglyphic symbols here and there in place of the more ordinary ones. Most occur in the last 20 columns of the inscription's 222, in a section recording the monuments that Khnumhotep had erected in the service of the pharaoh Amenemhet II.

The intention was not to make it hard to read the text. It was to impart a dignity and authority to it, perhaps in the same way that a government proclamation will spell out "In the year of Our Lord One thousand eight hundred and sixty three" instead of just writing "1863." The anonymous scribe may also have been demonstrating his knowledge for posterity. Thus the inscription was not secret writing, but it incorporated one of the essential elements of cryptography: a deliberate transformation of the writing. It is the oldest text known to do so.

The transformations occur in funerary formulas, in a hymn to Thoth, in a chapter of the Book of the Dead, on the sarcophagus of the pharaoh Seti I, in royal titles displayed in Luxor, on the architrave of the Temple of Luxor, on stele, in laudatory biographic inscriptions. There is nothing meant to be concealed in all this; indeed, many of the statements are repeated in ordinary form right next to the altered ones. Why, then, the transformations? Sometimes for essentially the same reason as in Khnumhotep's tomb: to impress the reader. Occasionally for a calligraphic or decorative effect; rarely, to indicate a contemporary pronunciation; perhaps even for a deliberate archaism as a reaction against foreign influence.

But many inscriptions are tintured, for the first time, with the second essential for cryptology—secrecy. In a few cases, the secrecy was intended to increase the mystery and hence the arcane magical powers of certain religious texts. But the secrecy in many more cases resulted from the understandable desire of the Egyptians to have passersby read their epitaphs and so confer upon the departed the blessings written therein.

In Egypt, with its concentration upon the afterlife, the number of these inscriptions soon proliferated to such an extent that the attention and the goodwill of visitors flagged. To revive their interest, the scribes deliberately made the inscriptions a bit obscure. They introduced the cryptographic signs to catch the reader's eye, make him wonder, and tempt him into unriddling them—and so into reading the blessings. It was a sort of Madison Avenue technique in the Valley of the Kings. But the technique failed utterly. Instead of interesting the readers, it evidently destroyed even the slightest desire to read the epitaphs, for soon after the funerary cryptography was begun, it was abandoned.

7 Conclusions

The laboratory demonstrated how an English monoalphabetic substitution can be recovered without exhaustive key search. Frequencies suggested the initial **e** and **t** replacements; common patterns such as **the**, **that**, and **to** then constrained the solution. As the passage became readable, longer phrases and specific words resolved the rarer letters.

The 19 recorded steps recovered all 25 symbols present in the 2,567-letter cryptogram. The remaining alphabet pair was inferred by elimination and explicitly distinguished from observed evidence. Each recovered pair was applied consistently throughout the passage.

The central lesson is that frequency analysis is a starting point, not an automatic letter-ranking substitution. Statistical clues become convincing when supported by grammar, word shapes, and repeated occurrences. A large nominal key space does not protect a cipher that retains so much language structure. The exercise also connects with Laboratory Work No. 1: permuting an alphabet may enlarge the key space, but a fixed monoalphabetic substitution still preserves frequencies.

Report materials

This laboratory required no application source code. Its materials consist of the ciphertext, recovered key, progress samples, screenshots, and this report, kept under **Lab_2** in the laboratory repository:

<https://github.com/AverageCEnj0yer/Cryptography>

References

- [1] *Laboratory Work No. 2: Cryptanalysis of Monoalphabetic Ciphers*. Supplied course material, Sections 2.1–2.2. Theory and reference frequencies translated from Romanian for this report.
- [2] Daniel Rodriguez-Clark, Crypto Corner. *Frequency Analysis: Breaking the Code*. Interactive activity used for the laboratory. Accessed 9 October 2026.
<https://crypto.interactive-maths.com/frequency-analysis-breaking-the-code.html>
- [3] Gabriel Balan. *Laboratory Work No. 2 working records*. Supplied ciphertext, four frequency screenshots, ordered substitution notes, and intermediate plaintext extracts.

A Complete supplied ciphertext

The following transcription preserves the supplied letters, punctuation, line breaks, and missing spaces. Backslashes used to mark line breaks in the pasted message are formatting markers and are not part of the cryptogram. Case is ignored during frequency counting and substitution.

NG T OTF gvtisf 4,000 fvtip tjn, xg t wnrq htssvo Zvgvw LqdcdanioviXgj wqv wqXg ixaang nc
↪ wqv
Gxsv, t ztpwvi phixav plvwhqvo ndw wqvqxvinjsfuqp wqtW wnso wqv pwnif nc qxp snio'p sxcv-
tgo xg pn onxgj qvnuvgvo wqv ivhniovo qxpwnif nc hifuwnsnjf. Qxp rtp gnw t pfpwvz nc
pvhivwrixwXgj tp wqv znovig rniso lgnrp xw; qv dpvo gn cdssf ovkvsnuvo hnov nc
qxvinjsfuqxhpzfzans pdapwxwdwxngp. Qxp xgphixuwXng, htikvo tandw 1900 A.H. xgwn wqvsxkXgj
inhI xg wqv ztxg hqtzavi nc wqv wnza nc wqv gnasvztgLqgdzqnwvu XX, zvisf dpvp pnzv dgdpdts
qxvinjsfuqxh pfzansp qvvtgo wqviv xg usthv nc wqv zniv nioxgtif ngvp. Znpw nhhdi xg wqv
↪ stpw
20hnsdzgp nc wqv xgphixuwXng'p 222, xg t pvhwXng ivhniOXgj wqv zngdzvgwPwqtW Lqgdzqnwvu
qto vivhwvo xg wqv pvikXhv nc wqv uqtitnq TzvgvzqvwXX. Wqv xgwgwXng rtp gnw wn ztlv xw
qtio wn ivto wqv wvyw. Xw rtp wnXzutiW t oxjgXwf tgo tdwqnixwf wn xw, uviqtup xg wqv ptzv
↪ rtf
wqtW tjnkvigzvgw uinhstztwXng rxss puvss ndw "Xg wqv fvti nc Ndi Snio Ngvwqndptgo vxjqw
qdgoivo tgo pxywf wqivv" xgpwvto nc edpw rixwXgj "1863."Wqv tngfzndp phixav ztf tsnp qtkv
avvg ovzngpwtwXgj xqp lgnrsvojvni unpwvixwf. Wqdp wqv xgphixuwXng rtp gnw pvhiw
rixwXgj, adw xwxghniunitwvo ngv nc wqv vppvgwXts vszvvgwP nc hifuwnjituqf: t
ovsxavitwWitgpcniztwXng nc wqv rixwXgj. Xw xp wqv nsovpw wvyw lgnrg wn on pn.Wqv
witgpcniztwXngp nhhdi xg cdgvitf cnizdstp, xg t qfzg wn Wqnwq,Xg t hqtuwvi nc wqv AnnI nc
↪ wqv
Ovto, ng wqv ptihnuqtjdp nc wqv uqtitnqPvwX X, xg infts wxwsvp oxpustfvo xg Sdyni, ng wqv
tihqxwitkv nc wqv Wvzusv nc Sdyni, ng pwvsv, xgstdotwnif axnjituqxh xgphixuwXngp. Wqviv xp
gnwqXgj zvtgw wn avhnghtsvo xg tss wqxp; xgovvo, ztgf nc wqv pwtwvzvvgwP tiv ivuvtwvo
xgnioxgtif cniz ixjqw gvyw wn wqv tswvivo ngvp. Rqf, wqvq, wqvWitgpcniztwXngp? Pnzvwxzvp
cni vppvgwXtssf wqv ptzv ivtpng tp xgLqgdzqnwvu'p wnza: wn xzuivpp wqv ivtovi. Nhhtpxngtssf
cni thtssxjituqxh ni ovhnitwxkv vccvhw; itivsf, wn xgoXhtwv t hngwvzunitifuingdgXhtwXng;
↪ uviqtup
vkvg cni t ovsxavitwv tihqtxpz tp t ivthwXngtjtxgpw cnivXgj xgcsdvghv.Adw ztgf
↪ xgphixuwXngp tiv
wxghwdivo, cni wqv cxipw wxzv, rxwq wqvPvhngo vppvgwXts cni hifuwnsnjf-pvhivhf. Xg t cvr
htpvp, wqv pvhivhf rtpxgwgovvo wn xghivtpv wqv zfpwvif tgo qvghv wqv tihtgv ztjXhts unrvip
nchviwtXg ivsxjXndp wvywp. Adw wqv pvhivhf xg ztgf zniv htpvp ivpdswvo cinzwq
dgovipwtgotasv ovpxiv nc wqv VjfuwXtgP wn qtkv utppvipaf ivto wqvXivuxwtuqP tgo pn hngcvi
dung wqv ovutiwvo wqv asvppXgjp rixwwvg wqvivXg.Xg Vjfuw, rxwq xwp hngXvgwitwXng dung
wqv tcwvisxcv, wqv gdzavi nc wqvPvXgphixuwXngp pnng • uinsxcvitwvo wn pdhq tg vywvgw wqtW
wqv twwvgwXng tgowqv jnnorXss nc kxpwnip cstjjvo. Wn ivkXkv wqvxi xgwvivpw, wqv
phixavpovsxavitwvsf ztov wqv xgphixuwXngp t axw naphdiv. WqvF xgwinodhvo wqvHifuwnjituqxh
pxjgp wn htwhq wqv ivtovi'p vfv, ztlv qxz rngovi, tgowvzuw qxz xgwn dgixooSxgj wqvz - tgo pn
xgwn ivtoXgj wqv asvppXgjp. Xwrtp t pniw nc ZtoXpNg Tkvgdv wvhqgXbdv xg wqv Ktssvf nc wqv
Lxgjp. Adwwqv wvhqgXbdv ctXsvo dwwvisf. Xgpwvto nc xgwvivpwXgj wqv ivtovip, xwvXxovgwsf
ovpwinfvo vkvg wqv psxjqwvpw ovpxiv wn ivto wqv vuxwtuqP, cnipng tcwvi wqv cdgvitf
hifuwnjituqf rtp avjdg, xw rtp tatgongvo.

B Literal decrypted text

This is the direct character-for-character result of the recovered mapping, with plaintext letters in lowercase. Unlike the reading version in Section 6.3, it retains joined words, original line breaks, and the stray bullet. It provides an exact basis for re-encryption after ignoring ciphertext case.

on a day nearly 4,000 years ago, in a town called menet khufubordering the thin ribbon of
↳ the
nile, a master scribe sketched out thehieroglyphs that told the story of his lord's life—
and in so doing heopened the recorded history of cryptology. his was not a system of
secretwriting as the modern world knows it; he used no fully developed code of
hieroglyphicsymbol substitutions. his inscription, carved about 1900 b.c. into theliving
rock in the main chamber of the tomb of the noblemankhnumhotep ii, merely uses some unusual
hieroglyphic symbols hereand there in place of the more ordinary ones. most occur in the
↳ last
20columns of the inscription's 222, in a section recording the monumentsthat khnumhotep
had erected in the service of the pharaoh amenemhetii. the intention was not to make it
hard to read the text. it was toimpart a dignity and authority to it, perhaps in the same
↳ way
that agovernment proclamation will spell out "in the year of our lord onethousand eight
hundred and sixty three" instead of just writing "1863."the anonymous scribe may also have
been demonstrating his knowledgefor posterity. thus the inscription was not secret
writing, but itincorporated one of the essential elements of cryptography: a
deliberatetransformation of the writing. it is the oldest text known to do so.the
transformations occur in funerary formulas, in a hymn to thoth,in a chapter of the book of
↳ the
dead, on the sarcophagus of the pharaohseti i, in royal titles displayed in luxor, on the
architrave of the temple of luxor, on stele, inlaudatory biographic inscriptions. there is
nothing meant to beconcealed in all this; indeed, many of the statements are repeated
inordinary form right next to the altered ones. why, then, thetransformations? sometimes
for essentially the same reason as inkhnumhotep's tomb: to impress the reader. occasionally
for acalligraphic or decorative effect; rarely, to indicate a contemporarypronunciation;
↳ perhaps
even for a deliberate archaism as a reactionagainst foreign influence.but many
↳ inscriptions are
tinctured, for the first time, with thesecond essential for cryptology—secrecy. in a few
cases, the secrecy wasintended to increase the mystery and hence the arcane magical powers
ofcertain religious texts. but the secrecy in many more cases resulted fromthe
understandable desire of the egyptians to have passersby read theirepitaphs and so confer
upon the departed the blessings written therein.in egypt, with its concentration upon
the afterlife, the number of theseinscriptions soon • proliferated to such an extent that
the attention andthe goodwill of visitors flagged. to revive their interest, the
scribesdeliberately made the inscriptions a bit obscure. they introduced thecryptographic
signs to catch the reader's eye, make him wonder, andtempt him into unriddling them — and so
into reading the blessings. itwas a sort of madison avenue technique in the valley of the
kings. butthe technique failed utterly. instead of interesting the readers, itevidently
destroyed even the slightest desire to read the epitaphs, forsoon after the funerary
cryptography was begun, it was abandoned.